

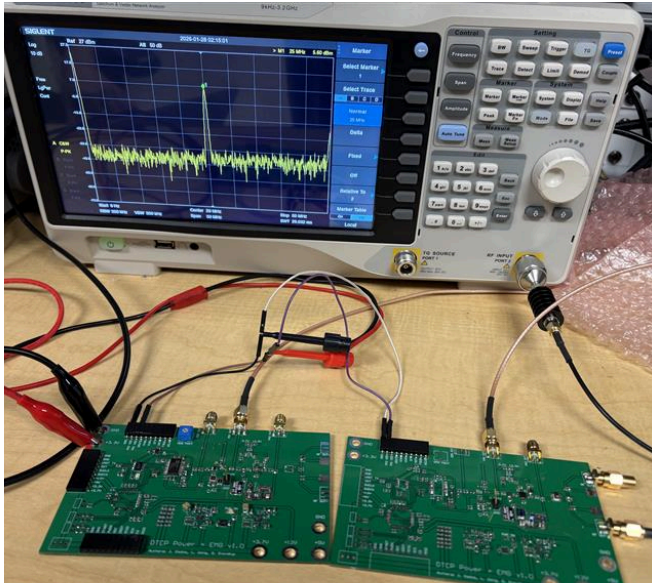
Alpha Build

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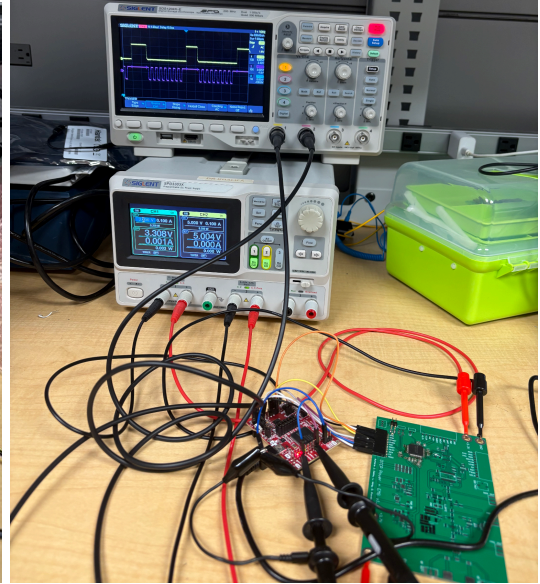
Project repository: <https://github.com/jdados/dtcp>

1. Current state of the project work

As of now, we have built a fully operational version of the system based on the PCB designed and ordered last semester. Currently, the system is split between 3 separate boards connected by jumper wires, as sub-systems were disconnected from each other to aid initial assembly and debugging. Nonetheless, the lab setups we have represent a vertical slice of all planned project features:



Setup 1



Setup 2

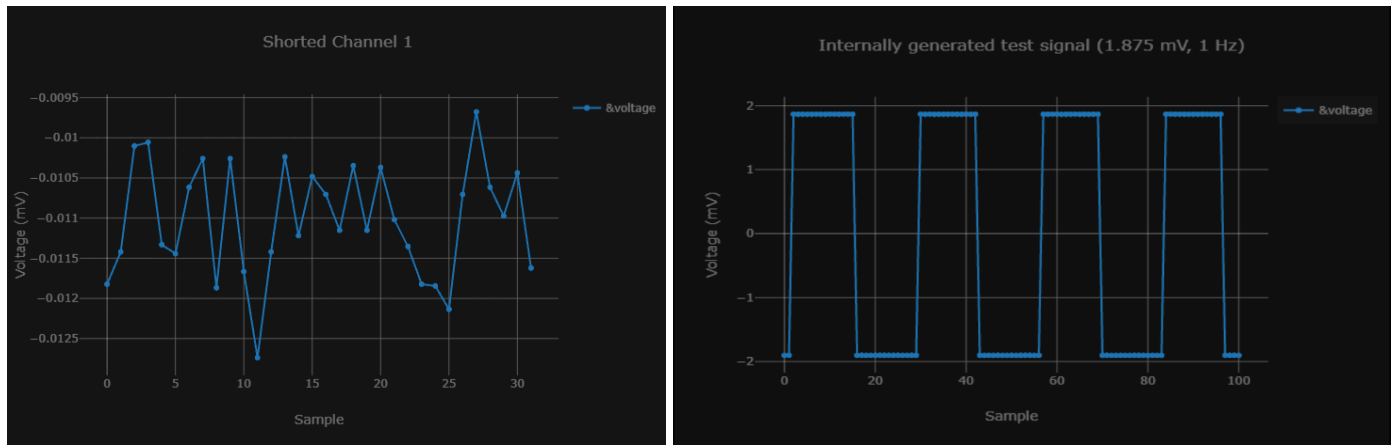
Setup 1 consists of two boards powered from a DC power supply. The left one contains the MCU, DDS, and the first-stage PA, and the right one contains the second-stage PA. The RF outputs are connected to a spectrum analyzer to validate the frequency and power.



The most important functionality (>30 dBm output) was confirmed using a Spectrum Analyzer (here with a 20 dB attenuator in series, bringing the signal from 30.67 dBm down to 10.67 dBm)

Setup 2 is used for EMG firmware development. It consists of a Launchpad with our MCU and debugger, connected to our custom PCB that contains only the AFE circuit. Input EMG signals are supplied by a signal

generator. Currently, we have tested the noise floor readings and test signal readings using the Launchpad debugger plotting the readings from memory:



Initial AFE readings measuring noise floor and a test signal.

2. Usability

a. Interface

The current version of the system is composed of several partially assembled boards connected together for debugging. The MCU part of the device is interfaced with the Launchpad, which provides firmware flashing and debugging functionalities. We are also developing a UART-based interface for data acquisition.

b. Navigation and perception

Once the device is flashed with firmware, the only controls required are the power supply connections and the output power setting potentiometer. Then, the board will start transmitting power to any 50 ohm load and performing the EMG measurements.

c. Responsiveness

As of now, the firmware is not meant to be interactive and simply performs voltage readings upon device initialization. In the future, LEDs will be utilized to show device status to the user.

3. Build quality

a. Robustness and Consistency

As of now, the system is stable for over at least 15 min of continuous operation and carries out RF transmission and EMG sampling without issues. There is some heating effect present while transmitting, and it will be quantified in future experiments.

b. Aesthetic rigor

For this iteration of the design, board aesthetics were not in-scope as the ease of debugging was prioritized.

4. Vertical features

a. External interface

The interfaces established and tested with our setups are the USB-debugger-MCU digital programming interface, the signal-AFE-MCU EMG interface, and the MCU-DDS-PA-load RF output interface. All of them are carrying out all intended functionality.

b. Persistent State

Currently, EMG recording data storage is only implemented on the microcontroller. Over the following weeks, we'll implement data movement from the microcontroller to the PC over UART.

c. Internal Systems

As of now, the interactions between all board sub-systems have been tested, errors have been identified, and with design revisions, the next PCB designed will capture all required interactions without the need for any modifications.